

1 **Supplementary Information**

2 **Supplemental Methods**

3 Study Overview

4 We encouraged staff to obtain SICU admission peri-rectal swabs in the operating room just prior to SICU
5 admission whenever possible to avoid discomfort after surgery. For patients who did not have a peri-rectal
6 swab in the operating room, admission swabs were obtained within 24 hours of admission to the SICU. For
7 consented and enrolled patients, the following information was collected: demographics, dates of
8 hospitalization, SICU admission and discharge dates, hospitalization and ICU stay within the last year,
9 transfer from an outside hospital, long-term hemodialysis status, in-hospital antibiotic exposure within 14
10 days prior to SICU admission, presence of an invasive device on SICU admission, surgical/invasive
11 procedure performed within 30 days prior to SICU stay, type of surgery leading to SICU admission, surgical
12 antibiotic prophylaxis, antibiotic exposure during SICU stay, device exposure, and repeat surgical/invasive
13 procedures during SICU and post-SICU hospital stay. The SICU includes a single hall with 11 beds arranged
14 side-by-side in two rows, four sinks dispersed between the beds, a nurse station with one sink, a nurse
15 breakroom, and a clean utility room (layout schematic in Supplementary Figure 1) which were sampled. Due
16 to budgetary and logistical constraints, only 5 of 11 bed rails and bedside cabinets were sampled, though
17 these locations represented each SICU region.

18 Reasons for not obtaining repeat swabs

19 The reasons for not obtaining repeat swabs at either SICU or hospital discharge were as follows: patient
20 discharged from hospital before research team approached (n=53 patients), rectal resection surgery (n=10),
21 patient declined (n=6), patient expired (n=5), and patients discharged within 24 hours of previous swab (n=2)

22 Microbial Culturing

23 Morphologically distinct isolates observed on MacConkey agar were subjected to individual testing. Species
24 identification and susceptibility testing of isolates presumed to be CRE were performed using the VITEK2
25 system (bioMérieux; Marcy-l'Étoile, France). Antimicrobial susceptibility testing (AST) was performed using
26 VITEK2 AST N406 card which includes 16 antibiotics with two carbapenems, imipenem and meropenem
27 (Supplementary Table 2). All isolates were frozen in tryptone soy broth with 15% glycerol at -40°C until
28 sequencing.

29 DNA Extraction and Whole-Genome Sequencing

30 For initial library preparation, 1µg of genomic DNA was fragmented, and adapters were attached to the
31 ends. Libraries were then size-selected and amplified using PCR with index primers, following the
32 manufacturer's guidelines. The PCR-amplified libraries were assessed for size and DNA quantity using the
33 TapeStation 4200 (Agilent Technologies). Subsequently, the paired-end library underwent sequencing on a
34 NovaSeq 6000 platform (Illumina, USA) at Unipath Specialty Laboratory Limited, Ahmedabad, India,
35 generating 2x150-bp reads. Reads with a PHRED quality score under 20 were discarded, and adapters
36 were removed using cutadapt¹ v1.8.1. The quality of the sequencing data was then evaluated using
37 MultiQC² v1.27.

38 Read Processing and Genomic Analysis

39 Adapter sequences were trimmed with Trimmomatic³ v0.39; forward and reverse reads were paired with
40 BBDuk⁴ v38.63 and *de novo* assembled into draft genomes using SPAdes⁵ v3.15.3. Isolates that had
41 genome completeness <90% or contamination >5%, measured by Quast⁶ v5.2.0 and CheckM⁷ v1.2.1, were
42 considered low quality and excluded from further genomic analyses. Isolates were decontaminated as
43 needed using the NCBI Foreign Contamination Screen.⁸ Isolate species were identified using Mash⁹ v2.3.
44 Multi-locus sequence types (STs) were assigned using mlst¹⁰ v2.22.1 using PubMLST¹¹ typing schemes.
45 Protein-coding sequences, annotated with Prokka¹² v1.14.6, were used as input for pangenome

investigation using Panaroo¹³ v1.2.10. Core genes present in $\geq 95\%$ of isolates were aligned and converted to an approximate maximum likelihood tree using RaxML¹⁴ v8.2.12 for phylogenetic visualization. To further quantify genetic relatedness between isolates, core single-nucleotide variants (SNVs) were called against the Panaroo-generated core gene alignment file using snp-dists¹⁵ v0.2.8. A threshold to define isolates belonging to the same lineage clusters was empirically identified using the distribution of core SNVs for each genus. Isolates were designated as members of the same strain cluster if their core gene SNV distance fell below the threshold of 20 using the distribution of all-vs-all core gene SNVs (Supplementary Figure 2) and had an overall coverage breadth of $\geq 98\%$. For visualization of lineage clusters, the connecting edges are weighted as $\left(\frac{1}{\text{Number of SNVs between isolates} + 1} \right) \times 10$. Plasmidfinder¹⁶ v2.1.6 was used to identify any putative plasmid contigs. Antibiotic resistance genes (ARGs) were identified using AMRFinderPlus¹⁷ v3.10.42. Pairwise comparisons of antibiotic resistance gene abundance were performed using the Wilcoxon rank-sum test with Bonferroni correction for multiple testing, implemented via the `compare_means()` function in the ggpubr package¹⁸ using R¹⁹ v4.3.0.

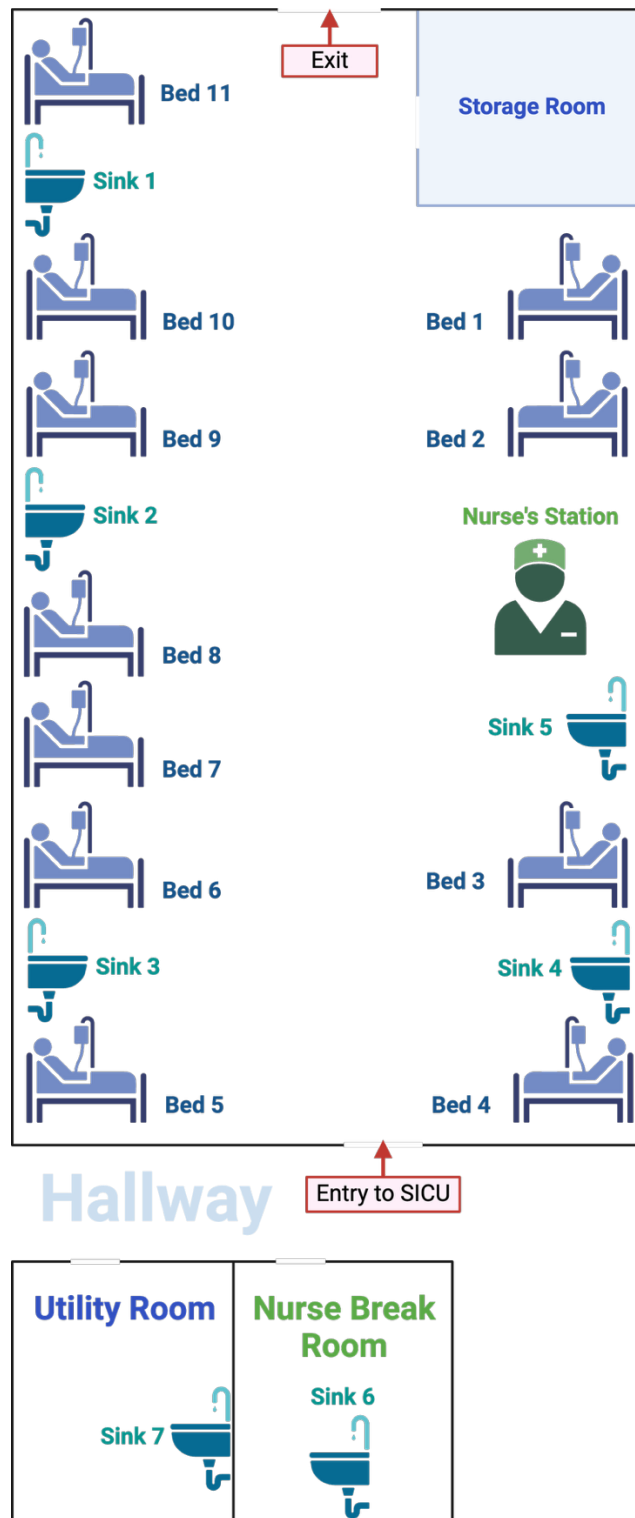
Risk Factor Analysis

Statistical comparisons were performed based on the determined origin of the CP-CRE isolate: community-associated (CA), healthcare-associated (HCA), hospital-acquired (HA), and environmental.²⁰ Univariate analysis was performed using Chi-Square and Fisher exact tests for categorical variables and the Kruskal-Wallis non-parametric test for continuous variables. Multivariable logistic regression analysis was performed on variables with $p < 0.05$ in univariate analysis using backward selection with likelihood ratios for examining the risk factors associated with CP-CRE peri-rectal colonization on SICU admission. Adjusted odds ratios (aORs) and 95% confidence intervals were calculated. Although Cox regression analysis was planned for examining the risk factors associated with CP-CRE acquisition, it was not performed as none of the variables were significantly associated in univariate analysis. All risk factor analyses were conducted using SPSS Statistics 27 software (IBM, Armonk, NY).

Supplementary Table 1: Risk factors associated with carbapenemase producing carbapenem-resistant Enterobacterales (CP-CRE) acquisition among surgical intensive care unit patients (SICU) during hospital stay in a tertiary care hospital, Kerala, India, July 29 – November 30, 2023

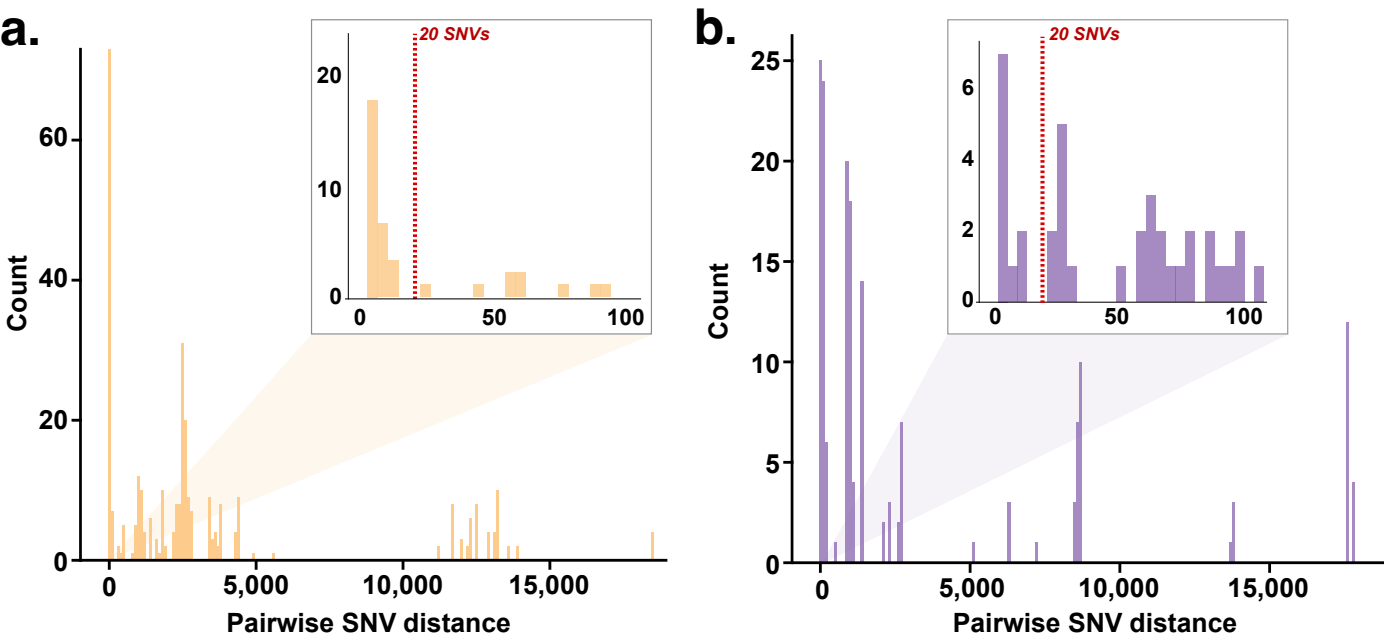
Risk factors	CP-CRE positive at hospital discharge N=42 (%)	CP-CRE negative at hospital discharge N=95 (%)	Univariable analysis* P-value
Age, median (IQR)	58 (46-68)	57 (45-67)	0.722
Female	20 (48)	37 (39)	0.342
Type of surgery			
Hepatobiliary surgery	12 (29)	29 (31)	0.818
Colorectal surgery	12 (29)	32 (34)	0.555
Upper GI and small intestinal surgery	9 (21)	22 (23)	0.824
Other surgery	8 (19)	10 (11)	0.181
Device exposure			
Urinary catheter	39 (93)	83 (87)	0.343
Central venous catheter	29 (69)	64 (67)	0.846
Arterial line	25 (60)	60 (63)	0.686
Mechanical ventilator	11 (26)	36 (38)	0.183
3 or more devices	20 (48)	40 (42)	0.549
New procedures/surgeries after initial surgery	9 (21)	12 (13)	0.205
Antibiotic exposure from SICU admission to hospital discharge			
Any antibiotic exposure	23 (55)	40 (42)	0.171
Two or more antibiotics exposure	10 (24)	16 (17)	0.352
Carbapenem exposure	2 (5)	7 (7)	0.544
Anaerobic agent exposure	20 (48)	36 (38)	0.201

*Univariate analysis was performed using Chi-square test and Fisher-exact test; Kruskal-Wallis non-parametric test was used to compare median values. All tests are two-sided.
CP-CRE: carbapenemase-producing carbapenem-resistant *Enterobacterales*; CI- confidence interval; HR- Hazard Ratio; ICU- intensive care unit; IQR: Interquartile range; SICU- surgical ICU



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82 **Supplementary Figure 1.** Overview of study SICU and sink locations. Figure generated with BioRender
 83 and available under agreement VB29ON9994.



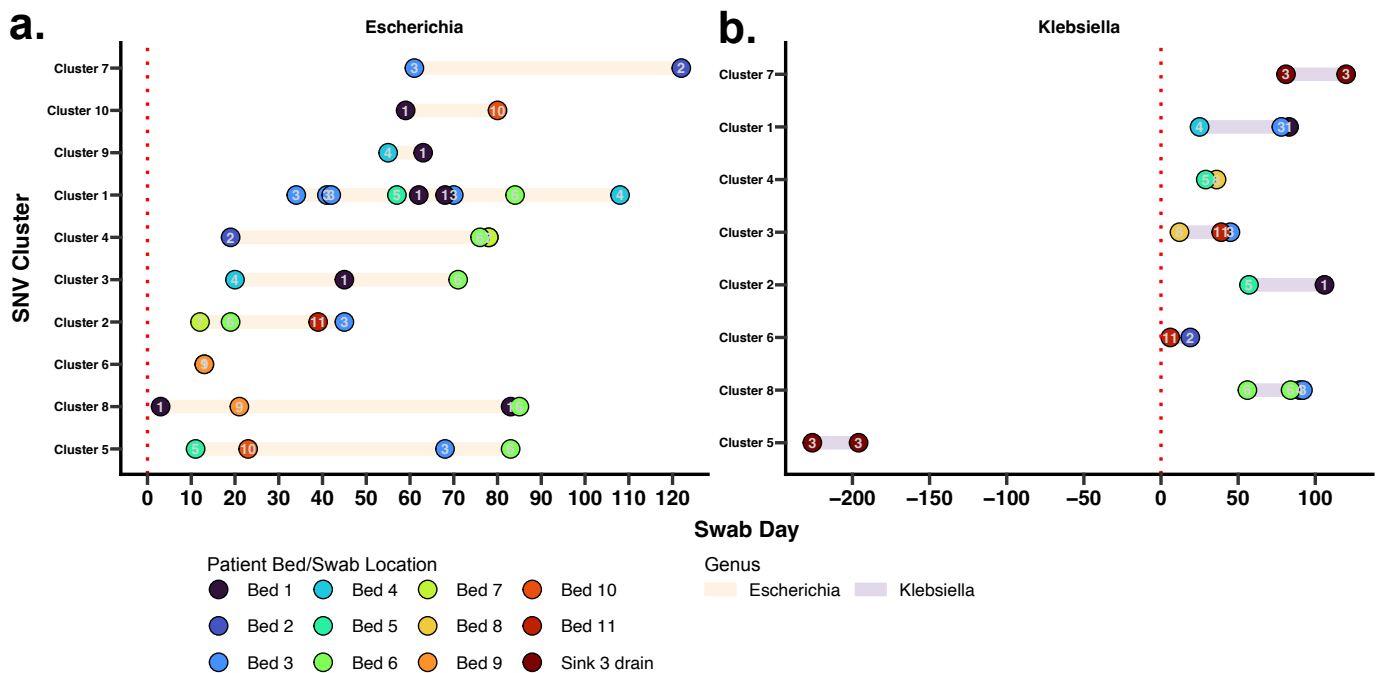
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86 **Supplementary Figure 2.** Core gene pairwise SNV distances among within-study isolates for **a** *E. coli* and

87 **b** *Klebsiella*. Source data are provided as a Source Data file.

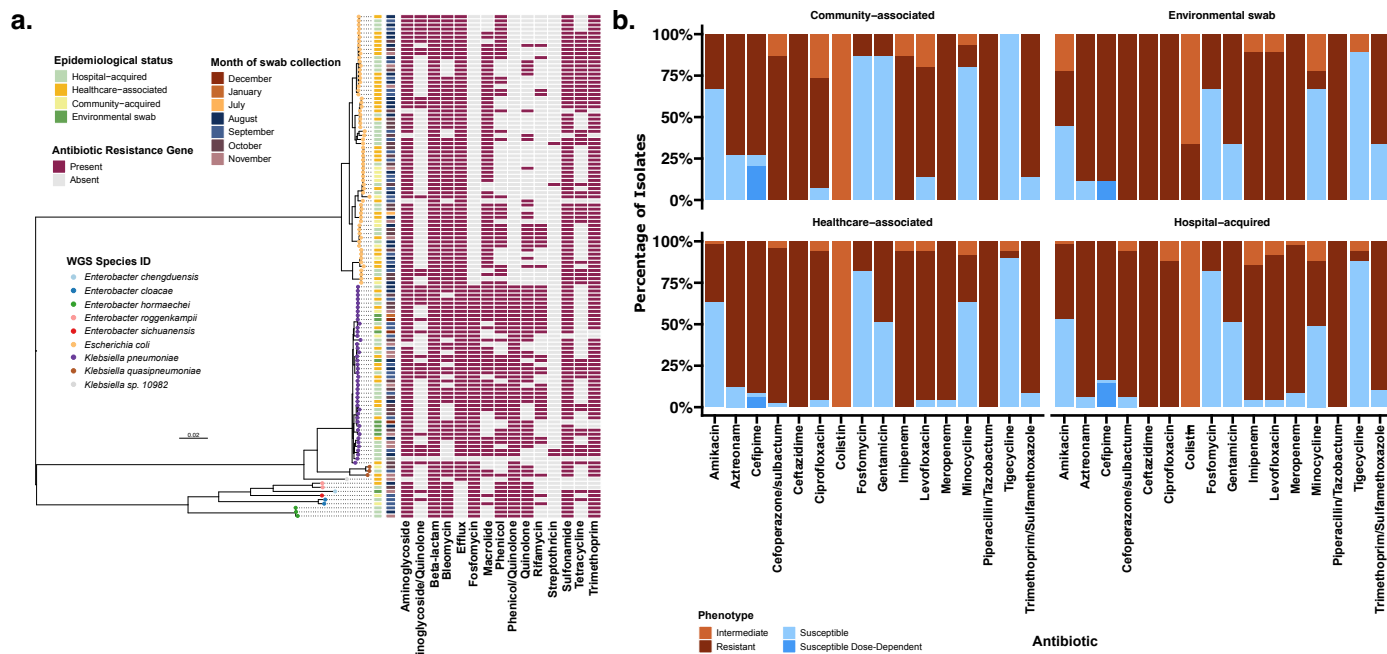
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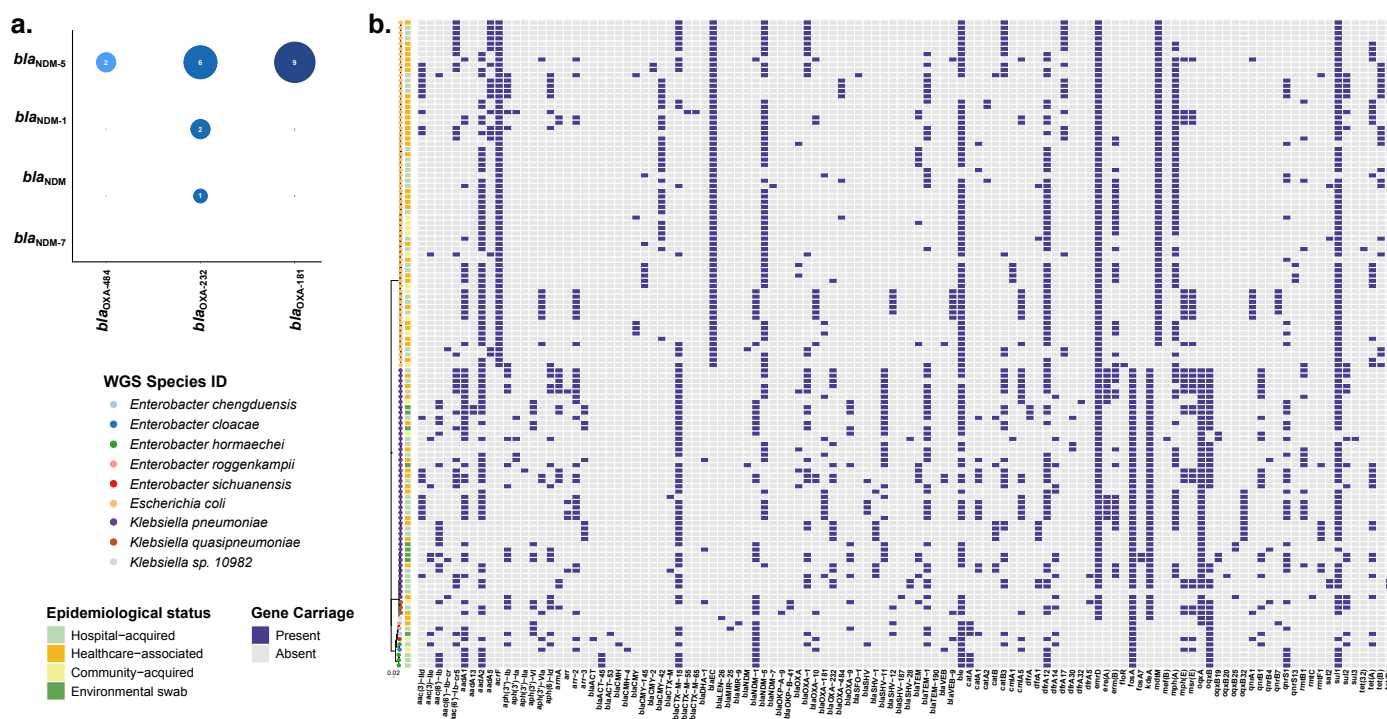


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91 **Supplementary Figure 3.** Core gene SNV-based clustering of **a** *E. coli* and **b** *Klebsiella* isolates. Unique
92 clusters include two or more isolates within <20 core gene SNVs of each other. Each node represents a
93 sample and is color-coded and labeled according to patient bed or environmental swab location. Samples
94 within the same cluster are arranged by the swab collection day. Day 0 is marked with a red dashed line and
95 denotes the first day of patient swab collection. Source data are provided as a Source Data file.

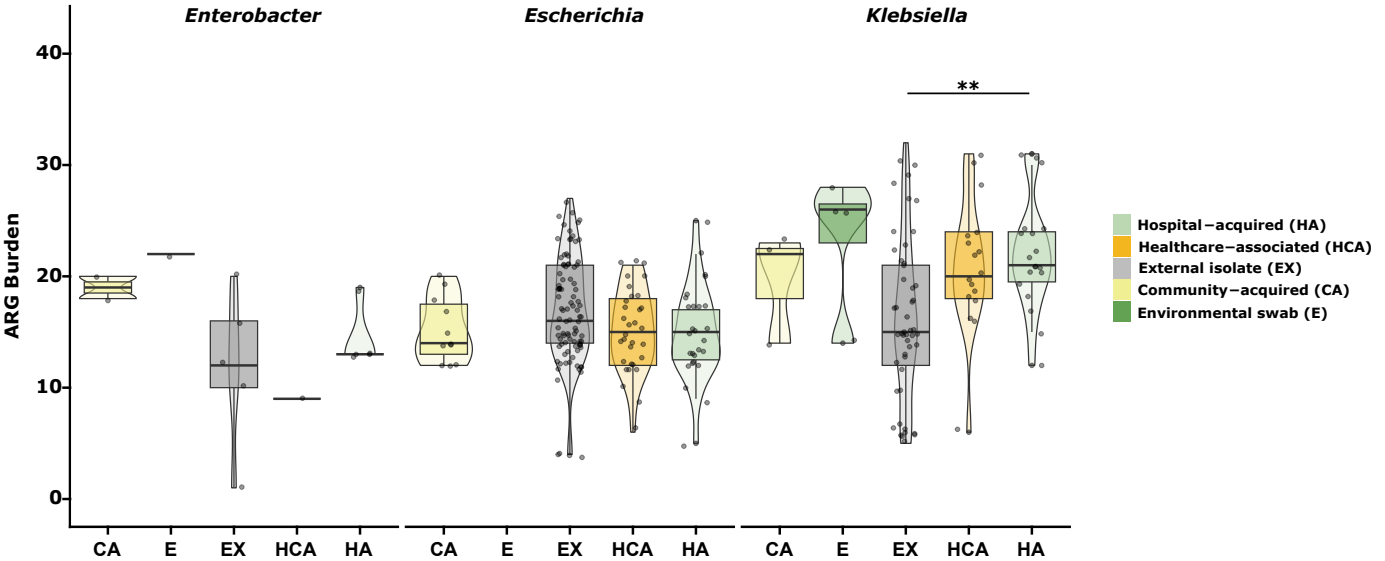
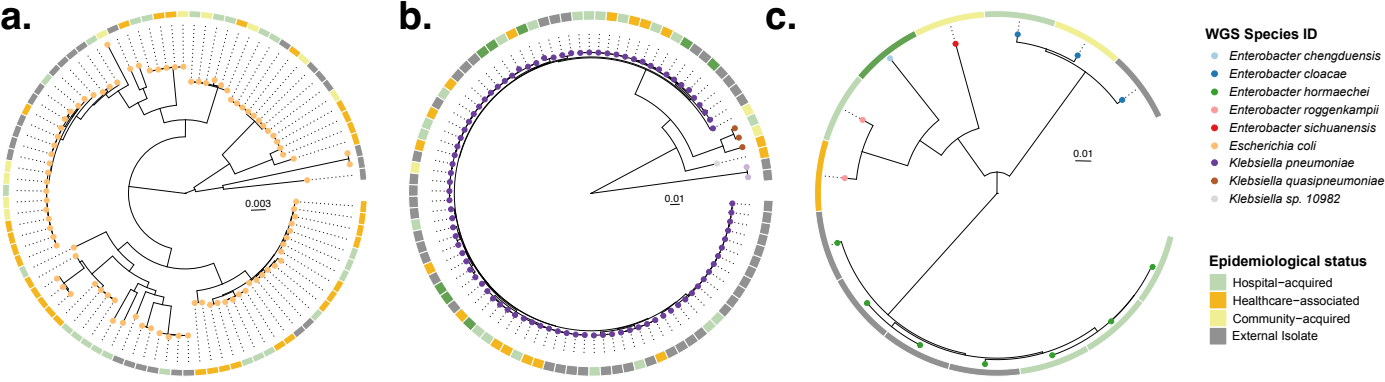


Supplementary Figure 4. **a** Maximum likelihood core gene phylogenetic tree of all CRESCENT isolates, annotated by species, epidemiological status, month of collection, and presence of genes corresponding to each antibiotic class. **b** Proportion of phenotypic resistance to varying antibiotics, stratified by epidemiological status.



Supplementary Figure 5. **a** Bubble plot illustrating the co-occurrence of NDM and OXA beta-lactamase genes carried by isolates. **b** Maximum likelihood core gene phylogenetic tree of all CRESCENT isolates, annotated by species, epidemiological status, and presence of antibiotic resistance genes and mutations sorted alphabetically. Source data are provided as a Source Data file.

112 **Supplementary Figure 6.** Maximum likelihood phylogenetic trees based on core genes of **a** *Escherichia*, **b**
113 *Klebsiella*, and **c** *Enterobacter* isolates. Isolates include samples from both this study and external isolates
114 listed in Supplementary Table 3. Trees are annotated with species identification and epidemiological status.
115 Source data are provided as a Source Data file.



118 **Supplementary Figure 7.** Box plots show the median and interquartile range (IQR) of antibiotic resistance
119 gene burden, stratified by epidemiological status for isolates from this study (n=123) and external isolates
120 (n=148) listed in Supplementary Table 3, with separate facets for each genus; whiskers denote values within
121 1.5× IQR. Individual data points are displayed as jittered points, and violin plots illustrate the kernel
122 probability density of the data. Differences between groups were assessed using a two-sided Wilcoxon rank-
123 sum test with Bonferroni correction ($p = 0.0021$). Significance levels are denoted as follows: $p < 0.05$ (*), p
124 < 0.01 (**), and $p < 0.001$ (***). Source data are provided as a Source Data file.

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